STAT6207 · FINAL GROUP PROJECT · ONE-PAGE PLAN

AI CV Matching Project

From a JD to a Top-5 shortlist: an explainable, evaluable matching baseline

A first-pass screening tool a senior recruiter would actually use

Second Talent
Student project briefing · 04 Nov 2026
v1.0 · One-page plan
Prepared for Eric Cheng · Chief of Staff

50	23	≥70%	4
tech good-match pairs (from 50 JD × 100 CV)	existing tech JDs (pilot scope)	target Top-5 hit rate	weeks from kick-off to delivery

01 Problem & objective WHY & WHAT

Second Talent fills **70–80%** of its roles in tech. Recruiters still match CVs by hand — slow, and inconsistent across reviewers. We will build a reproducible matching baseline for tech roles: **given one JD, return the Top-5 CVs with a short, evidence-based reason for each**, then test it against pairs a recruiter has approved.

Not a production feature. The goal is to prove that a simple, explainable matcher can meet a senior recruiter's shortlist bar.

Top-5 hit rate ≥ 70% · nDCG@5 ≥ 0.75 · Blind-review agreement ≥ 80%

02 Method & weights HOW IT WORKS

1. **Read JD** — Turn the role into a **must-have / nice-to-have** checklist
2. **Read CV** — Extract the same fields: skills, years, seniority, location, salary
3. **Filter** — Location / time zone / language / notice period as hard gates
4. **Score** — Weighted sum; missing fields count as **neutral 0.5**, not zero
5. **Rank** — Top-5, each with a **verifiable, evidence-based reason**

Tier	Pts	Factors
Highest	55	Hard-skill overlap **35** · Relevant years **20**
High	25	Seniority **15** · Salary vs budget **10**
Medium	20	Domain **10** · Stability **5** · Education / certs **5**
Filters	Gate	Location / remote · notice period · language → **pass / fail**

Tech route: v0 keywords → v1 BM25 + skill dictionary → v2 embedding hybrid. With only 50 pairs we will not train a complex model; ML is an ablation only. **Weights and targets freeze after Week 1 sign-off.**

OUTPUT EXAMPLE · JD: SENIOR BACKEND ENGINEER (ILLUSTRATIVE)

1. **CV-018 · 92** — 6/6 must-have skills; 7 yrs Go/Python; missing Kubernetes; location & salary fit
2. **CV-027 · 84** — 5/6 must-have skills; 5 yrs Python; microservices experience; salary to confirm

03 Evaluation HOW WE PROVE IT

- **Gold set:** rule-generated candidates from 50 JDs × 100 CVs, recruiter-confirmed 50 tech positives + 50 hard negatives; 0–3 rubric (3 = strong yes, 2 = shortlist, 1 = borderline, 0 = no).
- **Reliability:** first 20 pairs double-annotated; report Cohen's kappa; adjudicate disagreements; every label cites CV evidence.
- **Protocol:** leave-one-JD-out to avoid leakage; compare against random, keyword, and embedding baselines.
- **Metrics:** primary Top-5 hit rate and nDCG@5; secondary P@5, MRR, extraction F1, reason verifiability; error taxonomy: parsing / missed must-have / seniority / location-salary / hallucinated reason.

04 Data & scope DATA & SCOPE

Starting sheet	**50** JDs / **100** CV summaries (tech **23** / **46**)
Hired examples	**10** reference cases (rubric calibration)
Match labels	**0** now → **50** tech pos + **50** neg

Key gap: the brief targets 100–200 tech JDs; the starting sheet (*Manatal Examples…xlsx*) has 23; the 10 hired examples are reference only. This term: 23-JD pilot + request 30–50 more; else state scope and scaling plan in the report.

✓ In scope
Tech roles (AI/ML, Backend, Full-stack) · anonymised JD/CV · 50 labels · Claude extraction + baseline

✗ Out of scope
Unmasked data · live HRIS feature · buying/scraping LinkedIn · sales roles

05 4-week timeline TIMELINE

- **W1 Data & guide** — Index, align PDFs by Ref, schema, guide v1, calibrate rubric on 10 hired
- **W2 Labels & baseline** — Rule candidates; 50 tech pos + 50 neg + kappa; v0/v1; eval harness
- **W3 Enhance & evaluate** — Embedding hybrid, full evaluation, error analysis, reasons, blind review
- **W4 Deliver** — Report, demo, one-pager, buffer & rehearsal

06 Risks & mitigations RISKS

- **Only 23 tech JDs** → pilot first + request more; narrow scope if needed
- **100 labels is still small** → rules / embeddings; complex models only as ablation
- **Anonymised CVs lack salary/notice** → treat unknown as neutral 0.5, not a fail
- **Rule self-validation / subjective labels** → independent confirmation + leave-one-JD-out + double annotation + kappa
- **Privacy** → masked copies only, manual audit, Eric's approval before any data leaves

07 Deliverables DELIVERABLES

1. **Data pack:** jds/cvs JSONL, 50 positives + 50 negatives, data dictionary
2. **Matching guide (2–3 pp):** extraction schema, weights, 0–3 rubric
3. **Python baseline repo:** extraction, matching, evaluation, reproducible demo
4. **Evaluation report (5–8 pp):** metrics, error analysis, blind-review results
5. **Defense demo + one-page plan:** 10-minute walkthrough

Source: Second Talent · AI CV Matching Project Student Briefing (30 Oct 2026). **Privacy:** no data leaves Second Talent until masked, manually audited, and approved by Eric.

Team: _____
Members: _____